

Spiral of Silence:

How Neutral Moderation Polarizes Content Creation

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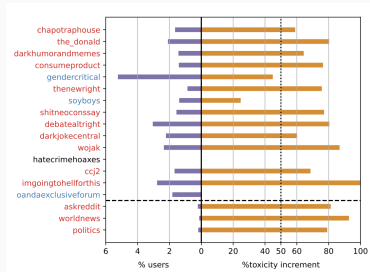
Online Toxicity – A Growing Concern

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- 71% support tighter platform rules on toxic content (Pew Research Center, 2021).

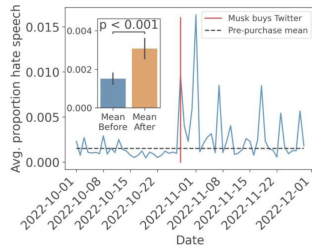
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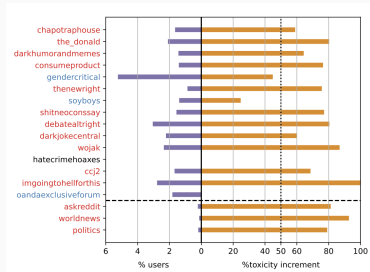
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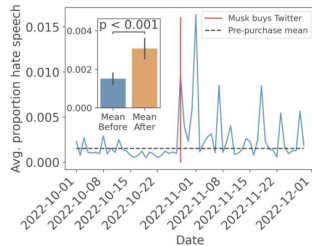
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- When Twitter **relaxed** its rules, hate speech **doubled** too.

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If both moderation and deregulation increases toxicity, then *what*, is being moderated?

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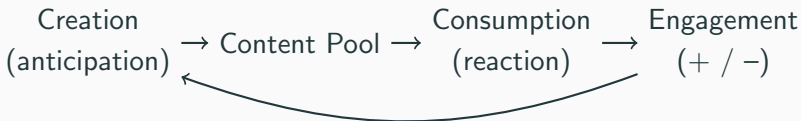
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- But moderation changes not only “who sees what”, but also *who* finds it worthwhile to **create**.

Our paper: From Reaction to Anticipation

- On social media, every user is **both** a consumer and a (potential) creator.
 - Creation *anticipates* engagement.
 - Consumption *reacts* to what survives moderation.

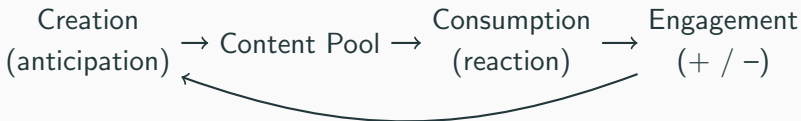
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- Our focus: Moderation shapes not only what is removed, but what is **never created**.

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- Even **neutral** policy can **polarize** content creation:
a structural *market failure* of social networks.

- Main Model.
- Extensions:
 - **Personalization**: ideology-based curation
 - **Neutrals**: indifferent to partisan ideology
 - **Targeted Toxicity**: toxicity itself becomes relational
- Conclusion and Implications

The Model

- A mass 1 of users differ in two dimensions $(i, t) \in \{A, B\} \times \{T, NT\}$:
 - ideology type $i \in \{A, B\}$ reflects their ideology.
 - toxicity type $t \in \{T, NT\}$ reflects their propensity to create toxic content.
- All users have a (potential) dual role as:
 - **readers** who read content created by others,
 - **creators** who post content that are read by others.

Content Consumers: Utility

A reader of type (i, t) derives utility U from consuming content created by a type (i', t') user:

$$U^r(i, i', t') = \underbrace{\alpha \cdot H(i, i')}_{\text{horizontal (ideology-based)}} + \underbrace{(1 - \alpha) \cdot V(t')}_{\text{vertical (toxicity-based)}} + \varepsilon_r$$

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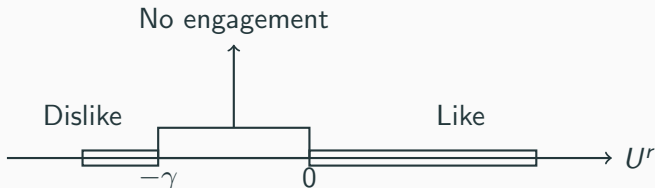
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- **horizontal** preference: $H(i, i') = \begin{cases} 0 & \text{if } i = i' \\ -1 & \text{if } i \neq i' \end{cases}$
- **vertical** preference: $V(t') = \begin{cases} 0 & \text{if } t' = NT \\ -1 & \text{if } t' = T \end{cases}$

Content Consumers: Engagement



- A reader i will **like** a post if $U^r(\cdot) \geq 0$.
- A reader i will **dislike** a post if $U^r(\cdot) < -\gamma$, where $\gamma \in [0, 1]$ calibrates the cost of dislike relative to that of likes.
 - Consistent with research (e.g. Stsiampkouskaya et al., 2023), users are more inclined to “like” rather than “dislike”
 - W.L.O.G. we normalize $\gamma = \frac{1}{2}$.

The hallmarks of **affective polarization** (α) (Iyengar, 2012):

- In-group *favoritism*: Readers react more **positively** to (even toxic) content from creators with the **same** ideology.
- Out-group *animosity*: Readers react more **negatively** to (even non-toxic) content from creators with **opposite** ideology.

Content Creators

- Given their type (i, t) , each user decides whether to create a post $a \in \{0, 1\}$. Utility from creation ($a = 1$) is given by

$$U^c(i, t) = \underbrace{S(t)}_{\text{survival rate}} \cdot \underbrace{R(V)}_{\text{content reach}} \cdot \underbrace{(NE^{in}(i, t) + NE^{out}(i, t))}_{\text{engagement}} \underbrace{+\epsilon_c}_{\text{intrinsic utility}}$$

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where

$$NE^{in}(i, t) = \underbrace{\lambda_i}_{\text{in-group share}} \cdot \left[\underbrace{P_l(i, it)}_{\text{prob. of like}} + \omega \underbrace{P_{dl}(i, it)}_{\text{prob. of dislike}} \right],$$
$$NE^{out}(i, t) = \underbrace{\lambda_{-i}}_{\text{out-group share}} \cdot [P_l(-i, it) + \omega P_{dl}(-i, it)],$$

- $\omega \in [-1, 0]$: the relative *weight* of negative over positive engagement on content creation (we relax it later).

► Extension: $\omega > 0$

Content Moderation and Reach

For analytical tractability, we model moderation intensity via a **survival** rate:

$$S(t) = 1 - \underbrace{\beta}_{\% \text{ toxic posts removed}} \cdot \mathbf{1}[t = T];$$

and content **reach** (from the creator's perspective) as:

$$R(V) = 1 - V; \quad V = \sum_{i,t} \underbrace{\lambda(i,t)}_{\text{share}} \cdot S(t) \cdot \underbrace{P_c(i,t)}_{\text{creation prob.}}$$

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- **Crowding-out effect** ($R'(\cdot) < 0$) : More content from others **reduces** the chance any *single* post is seen — creators compete for limited user attention.

Rational Expectation Equilibrium: In equilibrium, the expected share of each user type that creates content *equals* the probability of creation implied by their utility.

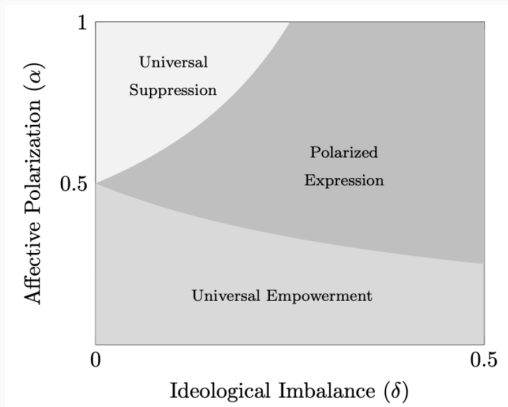
$$P_c(i, t) = Pr[U^c(i, t) > 0],$$
$$\forall i \in \{A, B\}, t \in \{T, NT\}$$

A unique solution exists by Brouwer fixed-point theorem.

Core Result: Spiral of Silence as Equilibrium

Proposition 1(c). Moderation Polarizes Creation: $\forall \beta \in [0, 1]$,

$$\alpha \in (\underline{\alpha}(\delta), \bar{\alpha}(\delta)) \Rightarrow \frac{\partial P_c^*(A, NT)}{\partial \beta} > 0 > \frac{\partial P_c^*(B, NT)}{\partial \beta},$$



Core Result: Spiral of Silence as Equilibrium

- Each takedown removes a toxic post, but also **reweights** *expectations* of in- vs. out-group exposure.
- This reweighting amplifies the *impact* of **asymmetry**:
 - **Ideological imbalance** (δ): unequal group sizes creates asymmetry in **influence**.
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- Moderation triggers a feedback loop:
 - The **majority**, sustained by in-group **favoritism**, creates more.
 - The **minority**, discouraged by out-group **animosity**, creates less.

Key Implication: The Appearance of Bias

- Critics call it “censorship”; empirical studies call it “toxicity asymmetry” (Haimson et al., 2021).
- Our model shows a third logic:
what looks like bias can endogenously *arise* as equilibrium **self-censorship**.
- The “spiral of silence” (Noelle-Neumann, 1974) is thus not enforced; but **induced** by moderation.

Average Wins Can Hide widening Gaps

- **Proposition 2.** Moderation **redistributes** reader welfare:

For all $\alpha > 0, \delta > 0, \beta > 0$, we have:

$$\frac{dE[\Delta U]}{d\beta} = \frac{dE[U^r(A) - U^r(B)]}{d\beta} > 0$$

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- Relational Externality: majority content gain higher reach without internalizing its **negative spillover** on the minority.

Extension 1: Content Personalization

We modify the effective content reach of in-group vs. out-group as:

$$U^c(i, t) = S(t) \cdot \left[R^{in} \cdot NE^{in}(i, t) + R^{out} \cdot NE^{out}(i, t) \right],$$

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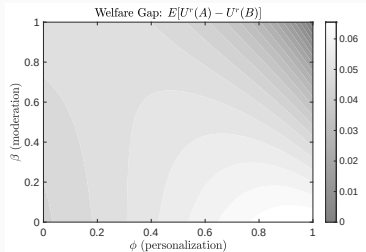
$$R^{in}(i, t) = R(\cdot), \quad R^{out}(i, t) = (1 - \phi) R(\cdot);$$

- $\phi \in [0, 1]$ captures how much personalization filters out cross-group exposure.

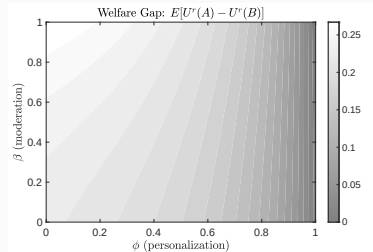
Personalization vs. Moderation

- Personalization is *not* a perfect complement for moderation:
 - **Conditional** relief on **non-toxic** creators (more in-group support, but also narrowed cross-group reach)
 - **Revive** minority **toxic** creation (via sympathetic in-group).
- Personalization could create a **subtler form of polarization** beyond echo chambers:
 - protects minority creation yet reshapes the content mix towards more toxicity;
 - even exacerbates “spiral of silence” at low α .

Personalization vs. Moderation



(a) $\alpha = 0.2, \delta = 0.1$



(b) $\alpha = 0.9, \delta = 0.1$

- Polarized spaces (e.g., X/Twitter): moderate personalization (ϕ) + strong moderation (β)
- Knowledge-based (e.g., Reddit/Wikipedia): too much personalization is counterproductive

Extension 2: Ideological Neutrals

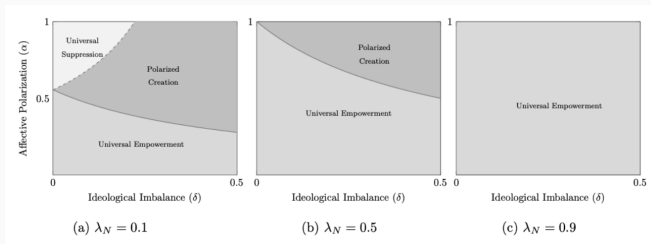
- Neutrals ($i = N$) do not care about ideology, only toxicity;
- Partisans ($i = A, B$) are more tolerable of neutral content:

$$H(i, i') = \begin{cases} 0, & \text{if } i = i', \\ -\frac{1}{2}, & \text{if } i' = N, \\ -1, & \text{otherwise.} \end{cases}$$

- Neutrals **dilute** the (affective) tension between partisans, but their creation also **crowds out** reach of partisan content.

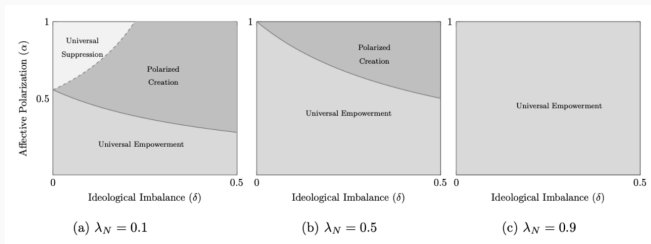
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- When neutrals form a **small minority**, stricter moderation can **deepen their own silence** despite declining toxicity on the platform (a parallel of “spiral of silence” among neutrals).

Extension 3: Targeted Toxicity Homophily

When toxicity itself becomes **relational**:

- Toxic readers value toxicity - when it targets the out-group.

$$U^r(r = it, c = i't') = \alpha H(i, i') + (1 - \alpha) T(it, i't') + \varepsilon_r,$$

$$T(it, i't') = \begin{cases} 0 & t' = NT \\ -1, & t' = T, t = NT \\ \frac{\kappa\alpha}{1-\alpha} \cdot \mathbf{1}[i = i'] - 1, & t' = T, t = T \end{cases}$$

where $\kappa \in (0, 1)$ captures the strength of such toxicity homophily.

- All qualitative results remain the same; but (surviving) toxic posts increases with κ .

Summary

- “Neutral” moderation is **not** neutral in outcome.
 - Moderation changes not just what gets removed, but
 - the relational environment that determines what gets *created* in equilibrium.
- Moderation generates content-level **externality**:
 - Affective polarization and ideological imbalance jointly create the asymmetry of emotional divide
- Regulating one dimension can distort the **supply** of another: toxicity (moderation) $\leftrightarrow$ ideology (personalization).

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– Karl Popper, Conjectures and Refutations (1963)

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Our model makes no claim to completeness, but it could be *useful*:

- **For policy:** suggests diagnose before intervention
 - Mislabeling disagreement as toxicity (because it's unpopular among the majority) could institutionalize “spiral of silence.”
- **For marketing:** reframes **affective polarization**, a core concept in political science, as a *structural feature* of social networks, which extends beyond political ideology per se (Lerman et al., 2024).